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Research Topic: Spectrally Emissive Radiative Cooling Roof Coating Optimized for Desert and Arid Climates.

Title: Spectrally Emissive Radiative Cooling Roof Coating Optimized for Desert and Arid Climates.

Introduction and Research Problem:

Buildings in the United Arab Emirates receive intense solar radiation for much of the year. Roofs face the sky directly and can become far hotter than the air, which increases conductive heat flow into occupied spaces. Air conditioning then removes this added heat. The Abu Dhabi Department of Energy reports that cooling accounts for about 60-70% of electricity use in UAE buildings [1]. A Dubai residential simulation also found that passive cooling measures can reduce annual energy use [2]. International guidance identifies reflective roofs and high-performance envelopes as practical ways to reduce cooling demand before mechanical systems are used [3], [4].

The roof is a major part of this problem. A hot-dry residential heat-gain study found that the roof contributed about half of the modeled heat gain while walls contributed about 30% [5]. Other envelope studies also show that roof reflectance and insulation strongly affect heat gain [6]. In Dubai, exposed roof surfaces can exceed 70 degrees Celsius during summer

afternoons [7]. These values vary by roof material and weather, but they show why the roof is an important target for passive cooling.

Passive daytime radiative cooling can reject heat without electricity. A suitable opaque coating reflects most sunlight in the 0.3-2.5 micrometer solar band and emits thermal radiation through the 8-13 micrometer atmospheric window. However, high reflectance alone does not guarantee strong field performance. Humidity absorbs outgoing infrared radiation, wind changes convective heat transfer, and desert dust can lower reflectance. A coating that performs well in a controlled test may therefore behave differently on a UAE roof.

The research gap is a controlled comparison that links measured solar reflectance and 8-13 micrometer emissivity to roof temperature and net cooling power under the same hot, dusty and variable outdoor conditions. This proposal extends my initial proof of concept into a replicated quantitative study. It will compare related coatings made from $BaSO_4$, rutile TiO_2, SiO_2 and an acrylic binder on identical panels. The study will identify which spectral property has the strongest effect during the day and at night and which formulation retains useful performance as humidity, wind and dust change.

Research Questions:

Central question: How do solar reflectance and mid-infrared emissivity in the 8-13 micrometer atmospheric window affect the net cooling power and surface temperature of a roof coating under UAE desert conditions with high solar irradiance, high air temperature and variable humidity and wind?

The study will answer five specific questions:

1. Thermal performance. Which coating produces the largest reduction in peak daytime surface temperature, nighttime temperature and test-box air temperature compared with conventional white roof paint?

2. Spectral relationship. How strongly do measured solar reflectance and 8-13 micrometer emissivity predict the observed temperature difference and net cooling power?

3. Weather sensitivity. How do humidity and wind speed change cooling performance during otherwise comparable periods?

4. Dust resistance. How much performance is lost after a controlled dust loading and how much is recovered after dry-brush cleaning?

5. Best formulation. Which formulation gives the best balance of cooling performance, environmental stability and practical application for a later roof-scale trial?

Literature Review:

Raman et al. demonstrated daytime radiative cooling with a photonic structure that reflected 97% of incoming sunlight and reached about 5 degrees Celsius below ambient under direct sun [8]. Mandal et al. later reported a scalable porous polymer coating with solar reflectance of 0.96 and long-wave infrared emittance of 0.97. It reached about 6 degrees Celsius below ambient and produced cooling power near 96 W per square meter under its test conditions [9]. These studies established that a daytime cooler must combine strong solar rejection with strong thermal emission.

Paint-format materials make this principle easier to apply to roofs. Li et al. developed a $BaSO_4$-acrylic paint with 98.1% solar reflectance and 0.95 emissivity in the sky window. It remained more than 4.5 degrees Celsius below ambient during field tests or produced an average cooling power of 117 W per square meter [10]. Dong et al. combined $BaSO_4$, $CaCO_3$ and SiO_2 particles to reach 97.6% solar reflectance. Their coating was 5.5 degrees Celsius cooler than commercial white paint during outdoor testing [11]. These results support the use of broad particle-size distributions and complementary materials rather than one pigment alone.

Environmental stability is a separate design problem. Hong et al. showed that fumed silica can reinforce a porous polymer matrix and preserve its radiative-cooling figure of merit when the coating is dried at relative humidity up to 60% [12]. Song et al. developed a TiO_2-based anti-aging paint whose solar reflectance changed by only 0.4% after an accelerated soiling test and 0.5% after simulated ultraviolet exposure [13]. Wang et al. also showed that an infrared-transparent polyethylene aerogel can reduce convective heat gain above a radiative coating [14]. A recent dual-layer roof study found large heat-flux and heat-gain reductions when phase-change materials were integrated into a hot-climate roof [15]. These approaches show that optical performance, convection and durability must be judged together.

The measurement plan was also informed by foundational and technical sources on heat transfer and solar irradiance. These include MIT OpenCourseWare [16], NASA guidance on total and spectral irradiance [17], a recent spectral-irradiance estimation study [18], calibration guidance for spectroradiometers [19] and practical explanations of irradiance measurement [20], [21]. These sources supported the terminology and instrument plan.

The performance targets and scientific comparisons in this proposal rely mainly on the peer-reviewed studies above.

Most published demonstrations optimize a material first and then test it under a limited set of weather conditions. Some field rigs shield the sample from wind to isolate radiation. A real roof is not shielded. The proposed work addresses this gap by measuring optical properties, surface temperature, enclosed-air temperature and weather data together. It compares five formulations on the same substrate and thickness range, quantifies humidity and wind effects and includes a controlled dust cycle. This climate-specific design is the main addition to existing work.

Proposed Methodology:

The method is a quantitative controlled experiment with replicated coating panels and repeated outdoor measurements.

1. Coating Formulation and Panel Preparation. Five groups will be prepared with three independent panels per group, giving 15 panels. Each panel will be a 10 cm by 10 cm aluminum sheet mounted over an identical insulated test box. The groups will be C0 conventional white acrylic roof paint, C1 rutile TiO_2 in acrylic, C2 $BaSO_4$ in acrylic, C3 $BaSO_4$ with SiO_2 in acrylic and C4 a multilayer system with a TiO_2-$BaSO_4$ reflective base plus a SiO_2-rich top layer. The substrate batch, surface preparation, binder system, panel area and curing period will be held constant. A drawdown bar will control the wet film. Dry thickness and coating mass per area will be recorded for every panel.

2. Optical and Thermal Characterization. Solar spectral reflectance will be measured from 0.3 to 2.5 micrometers with a UV-Vis-NIR spectrophotometer and integrating sphere. Infrared reflectance will be measured from 8 to 13 micrometers with an FTIR instrument. Because the coatings are opaque, emissivity will be calculated as one minus infrared reflectance. Three scans will be taken at different locations on each panel. A calibrated Type-K thermocouple will be fixed to the underside center of each panel with a thin layer of thermal paste. A second sensor inside each box will measure enclosed-air temperature. All temperature sensors will be compared on one shaded plate for 60 minutes. A sensor with a mean offset above 0.5 degrees Celsius will be corrected or replaced.

3. Outdoor Testing and Environmental Conditions. The panels will be mounted outdoors at the same fixed angle with equal sky exposure and at least 10 cm separation. A radiation-shielded sensor will record ambient temperature and relative humidity. An anemometer

will record wind speed and a pyranometer will record solar irradiance. All channels will be logged every minute and stored as five-minute averages for at least 14 summer days. Daytime analysis will focus on 10:00 AM to 4:00 PM and nighttime analysis on 9:00 PM to 2:00 AM. Panel positions will be randomized each morning. A separate wind test will expose each formulation to nominal air speeds of 0, 2 and 4 m/s for 30 minutes per condition on three clear days. Test order will be randomized.

4. Dust Resistance and Practical Evaluation. After the clean-panel phase, a measured mass of locally collected and sieved dust will be spread evenly on each panel. Surface loading will be recorded in grams per square meter. The panels will then be tested for three additional clear days. Reflectance will be measured before dust, after dust and after a standardized dry-brush cleaning. The study will report both performance loss and recovery. It will not claim long-term durability from this short cycle. Coating thickness, visible cracking, peeling and mass change will also be recorded as simple practical indicators.

5. Performance Evaluation and Recommendations. The primary outcome will be Delta T, defined as panel surface temperature minus ambient temperature. A negative value means sub-ambient cooling. Other outcomes will include peak daytime surface temperature, mean nighttime Delta T, maximum reduction relative to C0 and test-box air-temperature difference. Absorbed solar power will be calculated as (1 - Rsolar) multiplied by measured irradiance. Convective exchange will be estimated from the measured wind response instead of one fixed coefficient. A mixed-effects regression will test how reflectance, emissivity, irradiance, ambient temperature, humidity and wind predict Delta T and net cooling power. Panel and day will be random effects. Coating groups will also be compared by analysis of variance with Tukey comparisons. Dust performance will use paired clean-versus-dust comparisons. Results will report means, 95% confidence intervals, effect sizes and p values at a 0.05 significance level. Periods with rain, shading, sensor failure or missing irradiance will be excluded and reported. A Blender animation will explain the physics to a general audience, but it will not be treated as experimental evidence.

Expected Outcome:

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The multilayer coating is expected to approach solar reflectance above 0.96 and atmospheric-window emissivity above 0.94. Under clear low-wind conditions, it is expected to remain 6-12 degrees Celsius cooler than the conventional control during peak solar exposure and to reach sub-ambient temperature more often at night. These are

testable targets based on the initial project hypothesis and published performance ranges [8]-[11]. They are not presented as completed measurements.

The study is also expected to show that weather changes the value of spectral optimization. High humidity should reduce outgoing radiative heat loss while wind should pull a sub-ambient surface back toward air temperature through convection. Dust should reduce solar reflectance and weaken daytime cooling. The most useful result will therefore be a performance map that connects measured reflectance and emissivity with temperature, humidity, wind and dust condition. This is more informative than one peak temperature value.

Potential Limitations:

The main limitations are the small panel area, one outdoor location, a short summer test period and uncertainty in weather instruments. The dust cycle will show short-term sensitivity but not multi-year weathering. Test-box air temperature will indicate heat transfer but will not equal whole-building electricity savings. Optical instruments may also require access to an external laboratory. Replication, sensor calibration, recorded film thickness, randomized panel positions and weather-adjusted analysis will reduce these limits. If an integrating-sphere spectrophotometer or FTIR instrument is unavailable, the study will report that limitation and will not replace spectral measurements with unsupported estimates.

Conclusion:

This proposal tests a low-electricity response to a clear UAE building problem. Its novelty is not only the use of reflective and emissive particles. The project measures how those properties interact with the humidity, wind and dust that a real desert roof experiences. The five-stage design connects material formulation, spectral measurement, outdoor temperature data, environmental stress and statistical analysis. The result will be a defensible choice of coating for a larger roof trial rather than a claim based on one sample or one hot afternoon.

Project Practicalities:

The project can be completed in seven weeks. Week 1 will prepare substrates, complete pilot drawdowns and calibrate sensors. Week 2 will fabricate and cure the panels. Week 3 will complete spectral measurements and validate the rig. Weeks 4 and 5 will collect clean-panel field data plus the controlled wind tests. Week 6 will complete the dust cycle and repeat spectral measurements. Week 7 will verify calculations, prepare figures and finalize the report. Raw time-series data, calibration offsets, coating thickness and exclusion decisions will be retained in one traceable project folder.

Public-facing UAE sources on air-conditioning costs and household utility bills were part of the initial background research [22], [23]. They show the everyday relevance of cooling cost, but they are not used to set scientific performance targets. The experimental work requires only small panels, paint-format ingredients, low-voltage sensors and access to optical instruments. The design avoids human and animal participants.

Fine $BaSO_4$, TiO_2 and SiO_2 powders will be handled in small batches with a P2 or N95 respirator, nitrile gloves, long sleeves and sealed safety goggles in a ventilated enclosure. Safety data sheets will be reviewed before mixing. Guidance for coating powders identifies inhalation, skin contact and poorly ventilated spraying as important hazards [24], [25]. The study will avoid open flames, airborne powder clouds and compressed-air cleaning. Outdoor electronics will use low-voltage power, sealed connectors and shaded IP65 or higher enclosures. Cables will be secured to prevent trip hazards.

Post-program Plan:

If one formulation meets the optical and thermal targets, the next step will be a one-square-meter roof section tested beside a commercial cool-roof coating. A later study will monitor a full roof for at least one year and measure cooling-electricity use directly. Only after durability, waterproofing compatibility, cost and energy savings are verified should commercialization be considered. Upon completion, the researchers intend to refine the device through additional field validation and present the findings in academic conferences or publish them in a relevant journal.

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